

Section - B

Q: 2 (i) (or)

A: 2 (i) (or)

a) $x^2 - 4x - 5$

Solution :-

$$\begin{aligned}
 &= \cancel{10x^2} = x^2 - 5x + 1x - 5 \\
 &= x(x-5) - 1(x-5) \\
 &= (x-5)(x-1)
 \end{aligned}$$

b) $2x^2 + 9x + 4$

Solution :-

$$\begin{aligned}
 &= 2x^2 + 8x + 1x + 4 \\
 &= 2x(x+4) + 1(x+4) \\
 &= (2x+1)(x+4)
 \end{aligned}$$

Q: 2 (ii) (or)

A: 2 (ii) (or)

$8x^3 - 125y^3$

Solution :-

$$\begin{aligned}
 &= (2x)^3 - (5y)^3 \\
 &= (2x-5)(2x^2 + \cancel{2x}(5y) + 5y^2) \\
 &= (2x-5)(2x^2 + 10xy + 5y^2)
 \end{aligned}$$

Q: 2 (iii)

A: 2 (iii)

$12x^2y^3, 18xy^2$

$= 12 = 2 \times 2 \times 3$

$= 18 = 2 \times 3 \times 3$

$\Rightarrow 2 \times 3 = 6$

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$$x^2 = 20 \times 20$$

$$x = 20$$

$$y^3 = 9 \times 9 \times 9$$

$$y^2 = 9 \times 9$$

$$\text{HCF} = 6xy^2$$

$$\text{Q: } 2(iv)(or)$$

$$\text{A: } 2(iv)(or)$$

$$6a^2b \text{ and } 8ab^3$$

$$6, 8 = 24$$

$$a^2, a = a^2$$

$$b, b^3 = b^3$$

$$\text{L.C.M.} = 24a^2b^3$$

$$\text{Q: } 2(v)$$

$$\text{A: } 2(v)$$

$$4x^2 + 12xy + 9y^2$$

$$= (2x)^2 + 2(2x)(3y) + (3y)^2$$

$$= (2x + 3y)^2$$

$$= (2x + 3y)(2x + 3y)$$

$$= (2x + 3y)$$

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Q:2(vi)(or)
A:2(vi)(or)

$$5x - 2 - x = 4 - 3x - 27$$

Solution:-

$$\Rightarrow 5x - x + 3x = 4 + 2 - 27$$

$$\Rightarrow 4x + 3x = 6 - 27$$

$$\Rightarrow 7x = -21$$

$$\Rightarrow x = \frac{-21}{7}$$

$$x = -3$$

Q:2(vii)
A:2(vii)

$$|2x + 5| = 11$$

Solution:-

$$= 2x + 5 = 11, 2x + 5 = -11$$

$$2x = 11 - 5, 2x = -11 - 5$$

$$2x = 6, 2x = -16$$

$$x = \frac{6}{2}, x = \frac{-16}{2}$$

$$x = 3, x = -8$$

$$S.S = \{3, -8\}$$

Q: 2 (viii) (or)
A: 2 (viii) (or)

$$8x + 9 < 6x - 7$$

Solution::

$$8x - 6x < -7 - 9$$

$$2x < -16$$

$$x < \frac{-16}{2}$$

$$x < -8$$

$$S.S = \{x | x \in \mathbb{R}, x < -8\}$$

Q: 2 (ix)
A: 2 (ix)

$$A = (-2, 1)$$

$$B = (1, 5)$$

$$d_{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(1 - (-2))^2 + (5 - 1)^2}$$

$$= \sqrt{(1+2)^2 + (4)^2}$$

$$= \sqrt{(3)^2 + 16}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$= 5$$

$$\frac{\text{Q:2}(x)}{\text{A:2}(x)}$$

$$A := (2, 5)$$

$$B := (6, 9)$$

$$\text{formula} := \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}$$

$$\text{putting values} := \left(\frac{2+6}{2}, \frac{5+9}{2} \right)$$

$$= \left(\frac{8}{2}, \frac{14}{2} \right)$$

$$= (4, 7)$$

$$\frac{\text{Q:2}(x_i)}{\text{A:2}(x_i)}$$

$$A := (3, 2)$$

$$B := (4, 4)$$

$$C := (5, 6)$$

$$\begin{aligned} d_{AB} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(4 - 3)^2 + (4 - 2)^2} \\ &= \sqrt{(1)^2 + (2)^2} \\ &= \sqrt{1 + 4} \\ &= \sqrt{5} \end{aligned}$$

$$\begin{aligned} d_{BC} &= \sqrt{(5 - 4)^2 + (6 - 4)^2} \\ &= \sqrt{(1)^2 + (2)^2} \\ &= \sqrt{1 + 4} = \sqrt{5} \end{aligned}$$

$$\begin{aligned}
 d_{AC} &= \sqrt{(5-3)^2 + (6-2)^2} \\
 &= \sqrt{(2)^2 + (4)^2} \\
 &= \sqrt{4 + 16} \\
 &= \sqrt{20} \\
 &= 2\sqrt{5}
 \end{aligned}$$

Collinear ::

$$\begin{aligned}
 AB + BC &= AC \\
 \sqrt{5} + \sqrt{5} &= 2\sqrt{5} \\
 2\sqrt{5} &= 2\sqrt{5}
 \end{aligned}$$

Section - C

Q:3(i)

A:3(1)

$$(x+5)(x+3)(x+2)(x+6) - 88$$

Solution :-

$$\begin{aligned}
 &= [(x+5)(x+3)] [(x+2)(x+6)] - 88 \\
 &= [x(x+3) + 5(x+3)] [x(x+6) + 2(x+6)] - 88 \\
 &= [x^2 + 3x + 5x + 15] [x^2 + 6x + 2x + 12] - 88
 \end{aligned}$$

$$= [x^2 + 8x + 15] [x^2 + 8x + 12] - 88$$

$$= \text{let } x^2 + 8x = a$$

$$= [a + 15] [a + 12] - 88$$

$$= a(a + 12) + 15(a + 12) - 88$$

$$= a^2 + 12a + 15a + 180 - 88$$

$$= a^2 + 27a + 92$$

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$$\begin{aligned} &= a^2 + 23a + 4a + 92 \\ &= a(a + 23) + 4(a + 23) \\ &= (a + 23)(a + 4) \\ &= \text{putting the value back :-} \end{aligned}$$

$$(x^2 + 8x + 23)(x^2 + 8x + 4)$$

rearranging...

$$= (x^2 + 8x + 4)(x^2 + 8x + 23)$$

———— Q:3(ii) or ————

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Q:3(ii)

A:3(ii)

$$x^2 + 4x + 2$$

$$\begin{array}{r}
 x^2 \quad x^4 + 8x^3 + 20x^2 + 16x + 4 \\
 \underline{x^4} \\
 8x^3 + 20x^2 \\
 \underline{8x^3 + 16x^2} \\
 4x^2 + 16x + 4 \\
 \underline{4x^2 + 16x + 4} \\
 0
 \end{array}$$

Square root: $\pm (x^2 + 4x + 2)$

Q:3(iii)

$$2x + 3 > 7 \quad \text{OR} \quad 4x - 1 < 3$$

$$2x > 7 - 3 \quad \text{OR} \quad 4x < 3 + 1$$

$$2x > 4 \quad \text{OR} \quad 4x < 4$$

$$x > \frac{4}{2} \quad \text{OR} \quad x < \frac{4}{4}$$

$$x > 2 \quad \text{OR} \quad x < 1$$

$$S.S = \{x/x \in \mathbb{R}, x > 2 \text{ OR } x < 1\}$$

Q.3 (iv)

$$A: (-4, 2)$$

$$B: (1, 4)$$

$$C: (3, -1)$$

$$D: (-2, -3)$$

$$\begin{aligned}d_{AB} &= \sqrt{(1 - (-4))^2 + (4 - 2)^2} \\&= \sqrt{(1 + 4)^2 + (2)^2} \\&= \sqrt{(5)^2 + 4} \\&= \sqrt{25 + 4} \\&= \sqrt{29}\end{aligned}$$

$$\begin{aligned}d_{BC} &= \sqrt{(3 - 1)^2 + (-1 - 4)^2} \\&= \sqrt{(2)^2 + (-5)^2} \\&= \sqrt{4 + 25} \\&= \sqrt{29}\end{aligned}$$

$$\begin{aligned}d_{CD} &= \sqrt{(-2 - 3)^2 + (-3 - (-1))^2} \\&= \sqrt{(-5)^2 + (-3 + 1)^2} \\&= \sqrt{(-5)^2 + (-2)^2} \\&= \sqrt{25 + 4} \\&= \sqrt{29}\end{aligned}$$

$$\begin{aligned}d_{AD} &= \sqrt{(-2 - (-4))^2 + (-3 - 2)^2} \\&= \sqrt{(-2 + 4)^2 + (-5)^2} \\&= \sqrt{(2)^2 + 25} \\&= \sqrt{4 + 25} \\&= \sqrt{29}\end{aligned}$$

Diagonals:

$$\begin{aligned}d_{AC} &= \sqrt{(3 - (-1))^2 + (-1 - 2)^2} \\&= \sqrt{(3+1)^2 + (-3)^2} \\&= \sqrt{(7)^2 + 9} \\&= \sqrt{49 + 9} \\&= \sqrt{58}\end{aligned}$$

$$\begin{aligned}d_{BD} &= \sqrt{(-2 - 1)^2 + (-3 - 4)^2} \\&= \sqrt{(-3)^2 + (-7)^2} \\&= \sqrt{9 + 49} \\&= \sqrt{58}\end{aligned}$$

$$AC = BD = \sqrt{58}$$

All the vertices of this quadrilateral are equal to it is a square.

MCQ

1. A

2. A

3. A

4. C

5. A

6. A

7. C

8. A

9. B

10. B

11. B

12. C